Ring Artifacts Elimination Method in Computed Tomography using Sparsity-Induced Norms

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***Abstract*—Ring artifacts degrade the visibility of computed tomography images, where they appear as concentric circles in the reconstructed image. They are usually generated due to some limitations/failure associated with imaging hardware. Ring artifacts may destruct effective information in the image, and hence they should be carefully identified and eliminated without damaging other image structures. In this paper, we propose a ring artifacts elimination method that works within the image reconstruction process. With the existence of ring artifacts, the projection data can be represented as a sum of two components. The first contains perfect ideal projection data and the second contains imperfect elements causing the ring artifacts. We propose to minimize sparsity-induced norms such as a combination of ℓ_0 and ℓ_1-norms of the imperfect elements to remove the ring artifacts efficiently. The proposed algorithm is generated through optimizing the cost function with the alternative direction method of multipliers. The efficiency of the proposed method is tested qualitatively and quantitatively using simulation studies. Results of the reconstructed images show significant elimination of the artifacts.**

***Index Terms*—Computed tomography, image reconstruction, ring artifacts.**

I. Introduction

The quality of computed tomography (CT) images can be affected by the presence of different types of artifacts. Ring artifacts are considered a common type that frequently exists in several imaging applications. It presents as circles exist within the center of rotation in the reconstructed image. Ring artifacts could be generated if the detector is not correctly calibrated or have some defective elements (detector bins). Also, any damage caused to the scintillator screen in addition to beam instability can generate ring artifacts. Flat field correction is considered the typical technique in ring artifacts removal. Even though, some ring artifacts still appear after applying such technique. Moreover, different methods were investigated to eliminate the ring artifacts. For example, Munch *et al.*, investigated an integrated wavelet-Fourier filter (WFF) to remove the lines from the sinogram space [1]. Also, a technique focused on smooth generative adversarial networks was proposed for ring artifacts elimination in the Cartesian coordinates [2]. A correction approach was implemented in Cartesian and polar coordinate systems by employing mean and median filters, and they deduced that the correction approach in the polar coordinate system (RCP) produced higher accuracy [3]. In this study, we develop a new ring artifact elimination algorithm using sparsity-induced norms, which cannot be found in the existing literature. Our idea relies on separating the projection data into two components, the reliable data and the unreliable data that create the ring artifacts. We design the cost function in which sparsity-induced norms such as a combination of ℓ_0 and ℓ_1-norms for the unreliable elements and a smoothing penalty are minimized. Furthermore, we choose the alternative direction method of multiplier (ADMM) [4] to minimize the cost function.

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II. Proposed Method

The standard model for x-ray CT imaging in case of ring artifacts existence can be defined as follows:

$$A\vec{x} = \vec{b} + \vec{\lambda}, \tag{1}$$

where $\vec{b} = (b_1, \ldots, b_m)^\top$ is a vector for the reliable elements of the projection data, while $\vec{\lambda} = (\lambda_1, \ldots, \lambda_m)^\top$ is a vector for the unreliable elements in the projection data that cause ring artifacts and other noise effects. Also, $\vec{x} = (x_1, \ldots, x_n)^\top$ and $A = \{a_{ij}\}$ define vector for the image pixels and the $m \times n$ system matrix, respectively. From the physical model of ring artifact generation in CT, it is a reasonable assumption to consider $\vec{\lambda}$ as a sparse vector and its sparsity inspires us to use compressed sensing (CS) theory. The essential idea of this work is to exploit the CS to estimate the unreliable data in the projection data domain that generate the ring artifacts and remove them within the reconstruction. Consequently, we propose to optimize the following cost function by employing a combination of ℓ_0 and ℓ_1-norms for $\vec{\lambda}$ in the data fidelity as an example of the sparsity-induced norms to enforce the sparsity of unreliable component.

$$\text{Minimize}_{(\vec{x},\vec{\lambda})}\ \gamma\|\vec{\lambda}\|_0 + \gamma'\|\vec{\lambda}\|_1 + \beta\|\vec{x}\|_{TV} \text{ subject to } A\vec{x} - \vec{\lambda} = \vec{b}, \tag{2}$$

where $\|\vec{x}\|_{TV}$ is the total variation (TV), which can be denoted by

$$\|\vec{x}\|_{TV} = \sum_{s,t} \sqrt{(x_{s+1,t} - x_{s,t})^2 + (x_{s,t+1} - x_{s,t})^2}. \tag{3}$$

The augmented Lagrangian function corresponding to Eq. (2) is defined as:

$$L_\rho(\vec{x}, \vec{\lambda}, \vec{u}) = \gamma\|\vec{\lambda}\|_0 + \gamma'\|\vec{\lambda}\|_1 + \beta\|\vec{x}\|_{TV} + \frac{\rho}{2}\|\vec{\lambda} - A\vec{x} + \vec{b} + \vec{u}\|_2^2, \tag{4}$$

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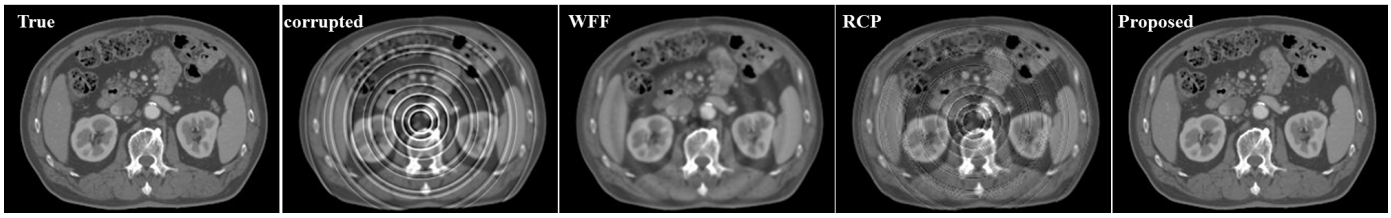

Fig. 1. Results of the proposed method and compared techniques. Display window: [0.7:1.5] cm^{-1}.

where both ρ and γ are positive parameters and $\gamma^{'} = 1 - \gamma$. Also, $\vec{u}$ defines the Lagrange multiplier. Equation (4) is optimized with ADMM method in which the following three steps are implemented for $k = 1, 2, \ldots, K$.

$$\vec{x}^{k+1} = \arg\min_{\vec{x}} \ \|\vec{x}\|_{TV} + \frac{\rho}{2}\|\vec{\lambda}^k - A\vec{x}^k + \vec{b} + \vec{u}^k\|_2^2. \quad (5)$$

$$\vec{\lambda}^{k+1} = \arg\min_{\vec{\lambda}} \ \gamma\|\vec{\lambda}\|_0 + \gamma^{'}\|\vec{\lambda}\|_1 + \frac{\rho}{2}\|\vec{\lambda}^k - A\vec{x}^{k+1} + \vec{b} + \vec{u}^k\|_2^2. \quad (6)$$

$$\vec{u}^{k+1} = \vec{u}^k + (\vec{\lambda}^{k+1} - A\vec{x}^{k+1} + \vec{b}). \quad (7)$$

The optimization of Eq. (5) can be made using proximal gradient descent algorithm. Hence, the image $\vec{x}$ will be updated as follows:

$$\vec{x}^{l+1} = \text{prox}_{\beta\|\vec{x}\|_{TV}}(\vec{x}^l + \alpha\rho A^T(\vec{\lambda} + \vec{b} + \vec{u} - A\vec{x}^l)), l = 1, 2, \ldots, L, \quad (8)$$

where L and α are the maximum iterations number for proximal gradient descent algorithm and the step size. The proximity operator regarding to the TV is similar to the minimization problem of TV denoising, which can be solved using standard method like Chambolle's projection algorithm [5]. Furthermore, we can observe that the optimization of Eq. (6) can be simply found, where it is considered as proximity operator of the summation of ℓ_0 and ℓ_1-norms.

III. Experimental Results

To evaluate the proposed method, we consider a slice of a 3D abdominal image obtained from medical image database. The slice size is 256 $\times$ 256 pixels and the projection data are acquired with 1000 views over 360° angles. To generate image contaminated by ring artifacts, we modified the sinogram by assuming that some random detector bins have a defect. Then, filtered back-projection algorithm is utilized to reconstruct the image. We compare the proposed algorithm with WFF technique [1] and RCP method [3]. The following parameters were selected for the proposed method and (WFF and RCP) techniques (L = 2, K = 500, $\alpha = 8.0\times10^{-6}$, $\rho = 0.1$, $\gamma = 0.5$, and $\beta = 5.0\times10^{-4}$) and (*wname*=44, L=3, and σ=0.6) and (T=0.5, M_r=7, and M_a=165), respectively. For fair comparison, parameters values of both WFF and RCP methods were chosen to achieve the highest image quality in terms of RRMSE values as well as the visual evaluation. Fig. 1 shows the results of the proposed algorithm and the compared techniques. Visually, it can be notable from the results that WFF technique can remove the ring artifacts, however it produces waves-like artifacts. Also, the rings are alleviated to some extend in the corrected images with RCP method but still exist. Whereas, we can observe that the reconstruction by the proposed algorithm can remove all the ring artifacts while retaining the image quality. To quantitatively evaluate the proposed technique, we calculated three image quality metrics: relative root mean square error (RRMSE), peak signal to noise ratio (PSNR) and structural similarity (SSIM). The measured values are displayed in Table I. It is clear that the reconstruction by the proposed method is superior to the compared methods applied in this work in terms of image quality metrics and elimination of ring artifacts.

TABLE I
THE VALUES OF THE MEASURED METRICS.

	RRMSE	PSNR	SSIM
WFF	0.076	32.37	0.8670
RCP	0.081	31.85	0.7833
Proposed	**0.005**	**55.07**	**0.9944**

IV. Conclusion

In this work, we proposed a method to remove ring artifacts from CT images during the image reconstruction procedure. The proposed method focused on minimizing sparsity-induced norms such as a combination of ℓ_0 and ℓ_1-norms for the unreliable values of the sinogram that produce ring artifacts. The performance evaluation of the proposed method is done using simulation data and compared with other competing techniques. The qualitative and quantitative results indicate that the proposed method is able to eliminate the ring artifacts and maintain the image details. The slow convergence of ADMM algorithm is considered a drawback of the proposed method. In the future, we plan to introduce faster optimization approach to overcome this drawback.

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